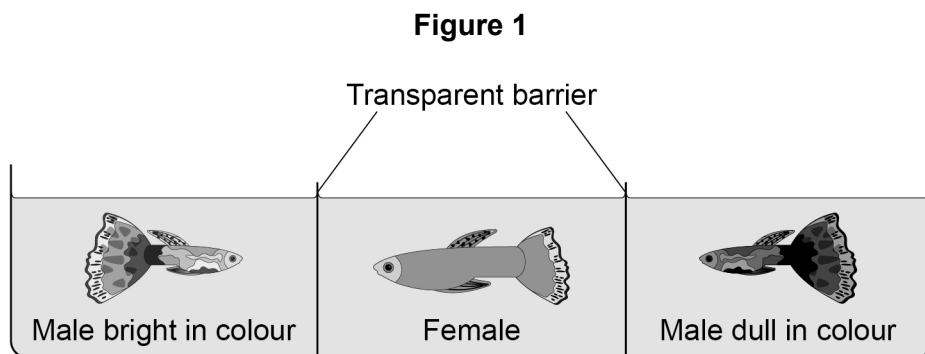


0 1

Guppies are small fish. Female guppies are dull in colour. Male guppies can be bright or dull in colour.

Scientists investigated the effect of female brain size on choosing a mate. They used laboratory-bred female guppies with large brains and with small brains.

They set up a fish tank as shown in **Figure 1**.



They observed each female for 10 minutes and recorded which male they were attracted towards. They repeated this with 45 large-brained females and 45 small-brained females.

0 1 . 1

Suggest **three** possible limitations of this investigation.

[3 marks]

- 1 _____

- 2 _____

- 3 _____

Guppies with large brains are better at identifying predators.

The scientists found that **only** female guppies with large brains were attracted to male guppies bright in colour.

0 1 . 2

Suggest and explain the advantage of this behaviour to the population of guppies.

[3 marks]

0 1 . 3

Describe how the behaviour of female guppies could result in sympatric speciation.

[3 marks]
